Chapter 9, Problem 1P

Problem

Given the sinusoidal voltage *v*(*t*) = 50 cos (30*t* + 10°) V, find: (a) the amplitude *Vm*, (b) the period *T*, (c) the frequency *f*, and (d) *v*(*t*) at t = 10 ms.

Step-by-step solution

Step 1 of 5

Consider the following data:

The sinusoidal voltage is as follows:

$v(t) = 50\cos(30t + 10°)$ ...... (1)

Consider the generalized voltage signal.

$v(t) = A\cos(\omega t + \phi)$ ...... (2)

Step 2 of 5

(a)

Calculate the amplitude, A

Compare the input sinusoidal voltage in equation (1) with generalized voltage in equation (2).

$A = 50$

Therefore, the amplitude of the sinusoidal voltage is $\boxed{50}$.

Step 3 of 5

(b)

Compare the input sinusoidal voltage in equation (1) with generalized voltage in equation (2).

$\omega = 30\text{ rad/s}$

Calculate the period, T

$$T = \frac{2\pi}{\omega}$$

Substitute 30 rad/s for ω in $T = \frac{2\pi}{\omega}$.

$$\begin{aligned} T &= \frac{2\pi}{30} \\ &= 0.2094 \\ &= 209.4\text{ ms} \end{aligned}$$

Therefore, the period of the sinusoidal signal is $\boxed{209.4\text{ ms}}$.

Step 4 of 5

(c)

Calculate the frequency, f

$$f = \frac{1}{T}$$

Substitute 209.4 ms for T in $f = \frac{1}{T}$.

$$\begin{aligned} f &= \frac{1}{209.4 \times 10^{-3}} \\ &= \frac{1000}{209.4} \\ &= 4.775\text{ Hz} \end{aligned}$$

Therefore, the frequency of the sinusoidal signal is $\boxed{4.775\text{ Hz}}$.

Step 5 of 5

(d)

Consider the following data:

The value of time, $t = 10\text{ ms}$

Calculate the sinusoidal voltage at $t = 10\text{ ms}$

Substitute 10 ms for t in $v(t) = 50\cos(30t + 10°)$.

$$\begin{aligned} v(t) &= 50\cos\left(30\left(10 \times 10^{-3}\right) + 10°\right) \\ &= 50\cos(0.3 + 10°) \end{aligned}$$

Here 0.3 is in rad. So, convert to degrees.

$$\begin{aligned} v(t) &= 50\cos\left(0.3\left(\frac{180}{\pi}\right) + 10°\right) \\ &= 50\cos(17.198° + 10°) \\ &= 50\cos(27.198°) \\ &= 44.475\text{ V} \end{aligned}$$

Therefore, the sinusoidal voltage $v(t)$ at $t = 10\text{ ms}$ is $\boxed{44.475\text{ V, } 0.3\text{ rad}}$.